

NIRIKSHAK (निरीक्षक) — 5-Minute Master Presentation Script (Hinglish)

Smart India Hackathon (SIH26034) | Ministry of Consumer Affairs, Food & Public Distribution

Department of Consumer Affairs (DoCA),
Government of India



Executive Presentation Metadata

- **Total Presentation Duration:** Exactly **5 Minutes** (300 Seconds)
 - **Language / Style:** Professional, Authoritative & Engaging **Hinglish** (Stage-ready)
 - **Target Pace:** 130–140 words per minute (Total spoken word budget: ~680–720 words)
 - **Design Philosophy:** *"AI Observes, Deterministic Rules Verify, Human Officer Decides."*
 - **Script Customization Feature:** Crucial lines include **[OPTION A]** (High Energy / Impact) and **[OPTION B]** (Formal / Gravitas) so the speaker can pick their preferred delivery style.
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Summary of Critical Corrections over Previous Script (**Script-1.pdf**)

Component	Previous Script (Script-1.pdf) Issue	Corrected & Verified Ground Truth	Why It Protects You in Front of Judges
Opening Hook	Unverified claim of "₹1,800 Crore+ per year revenue leakage due to font size or missing unit price".	Framed around 1.2 Crore retail establishments vs ~3,000 Legal Metrology Officers (1:4,000 deficit) and ₹25 Lakh Crore annual packaged consumer goods market .	No judge can challenge a phantom number; the 1:4,000 inspector deficit is official government data.
Inspection Timing	Inconsistent claims: stated <i>20–25 minutes</i> on page 1, but <i>8–10 minutes</i> on page 2.	Clarified: Comprehensive multi-panel inspection + Table-I font lookup + manual panchnama takes 20 to 25 minutes , compressed to under 2 minutes total officer workflow (with < 30s algorithmic processing).	Consistent numbers prevent judges from noticing conflicting math.
Court Evidence & Law	Claimed "35% <i>court dismissals under BSA 2023 Section 63</i> " without citation; claimed "100% <i>court admissibility & zero dismissals</i> ".	Stated the real statutory shift: 1 July 2024 repeal of Section 65B of Indian Evidence Act 1872 ; courts now demand cryptographic hash integrity under Section 63 of Bharatiya Sakshya Adhiniyam, 2023 (BSA) . Eliminates evidence tampering dismissals.	Proves deep legal research; avoids making legally impossible claims like "100% guaranteed judicial verdict".
Measurement Accuracy	Arbitrary <i>"±0.08mm sub- millimeter precision"</i> .	Cited verified empirical ground truth from real physical test: Himalaya Brahmi digital caliper truth = 1.47 mm ;	Real physical benchmark backed by code and photographs in the repository.

Component	Previous Script (Script-1.pdf) Issue	Corrected & Verified Ground Truth	Why It Protects You in Front of Judges
Nirikshak CV = 1.46 mm (0.01 mm delta!).			
Audience Tone	Mentioned <i>"And the best part for investors?"</i>	Replaced with "Department of Consumer Affairs aur State Governments ke exchequer ke liye" .	SIH evaluates solutions for the Government of India, not venture capitalists.
Proportional Justice	Ignored legal nuance; suggested direct penalties.	Integrated the Jan Vishwas Act, 2023 : 15-Day Improvement Notice for technical deficits vs ₹25,000 Civil Compounding for substantive violations.	Judges look for alignment with the Government's <i>Ease of Doing Business</i> mandate.

5-MINUTE TIME BUDGET BREAKDOWN			
SECTION	TARGET TIME	SPOKEN WORD COUNT	
PART 1: The Opening	0:00 - 1:15 (75s)	~170 words	
PART 2: The Core Tech	1:15 - 3:20 (125s)	~280 words	
PART 3: Feasibility & Impact	3:20 - 4:45 (85s)	~190 words	
Closing Punchline & Handoff	4:45 - 5:00 (15s)	~40 words	
TOTAL	5 Minutes (300s)	~680 words	



THE MASTER 5-MINUTE SCRIPT

PART 1: THE OPENING (Problem, Ground Reality & The NIRIKSHAK Thesis)

 **Allocated Time: 0:00 – 1:15 (75 Seconds)**

 **Objective:** Hook the judges immediately, expose the acute national enforcement bottleneck with verified government numbers, and state our core thesis.

[0:00 - 0:25] The Hook: National Scale vs Enforcement Bottleneck

(Stage Cue: Speaker steps forward, confident eye contact, clear deliberate voice)

[OPTION A - Problem-First Punchy Hook]:

*"Respected Judges, India mein har saal **₹25 Lakh Crore** se zyada ke pre-packaged consumer goods bikte hain — 1.2 Crore retail shops aur wholesale mandis ke through.*

*Lekin pure desh mein in arbon packages par Legal Metrology compliance verify karne ke liye hamare paas sirf **~3,000 Legal Metrology Officers** hain! Yani ek officer par chaar hazar se zyada establishments!"*

[OR OPTION B - Daily Citizen Perspective Hook]:

"Respected Judges, subah ke doodh aur biscuit se lekar life-saving dawaaiyon tak — hum rozana pre-packaged commodities par trust karte hain.

*Lekin 1.2 Crore retail dukanon ke samne pure desh mein sirf **~3,000 Legal Metrology Officers** hain. Ek inspector ke hisse 4,000 se zyada establishments aate hain, jisse 99% se zyada retail stock bina kisi inspection ke market mein circulate hota hai."*

[0:25 - 0:50] The Field Friction & The Legal Void

(Stage Cue: Hand gesture showing manual caliper / squinting at label)

"Lekin problem sirf manpower ki kami nahi hai, balki do aur acute bottlenecks hain:

Pehla: **Manual Friction**. Ek multi-panel carton ko vernier caliper aur magnifying glass se inspect karna, Principal Display Panel calculate karna, aur hath se panchnama likhna — ek officer ke **20 se 25 minute** le leta hai. Din bhar mein mushkil se 15 packages check ho paate hain.

Doosra: **The Evidentiary Void**. 1 July 2024 se Indian Evidence Act repeal ho chuka hai. Ab **Bharatiya Sakshya Adhiniyam (BSA) 2023 ki Section 63** ke tehat, smartphone se li gayi unverified digital photos court mein seedhe challenge ho jaati hain, kyunki evidence ki digital chain-of-custody missing hoti hai."

[0:50 - 1:15] The Introduction & Core Operating Thesis

(Stage Cue: Slide changes to Title & Architecture. Firm, authoritative delivery)

[OPTION A - Decisive & Architectural]:

"Namaste! Hum hain **Team NIRIKSHAK**, presenting an AI-Assisted Legal Metrology Inspection & Evidence Workstation for the Department of Consumer Affairs.

Hamara foundational golden rule non-negotiable hai:

'AI Observes, Deterministic Rules Verify, and the Human Officer Decides.'

Nirikshak 25-minute ke manual inspection ko **under 2 minutes** mein badalta hai — with sub-millimeter optical calibration aur court-admissible electronic evidence."

[OR OPTION B - Mission-Driven]:

"Namaste! We present **NIRIKSHAK** — jo 19th-century ke physical calipers ko 21st-century ki Calibrated Computer Vision aur Tamper-Proof Evidence se replace karta hai.

Hum AI ko judge nahi banate. Hamara architecture clear hai:

Computer Vision sirf aankh ka kaam karegi, faisla 100% Gazette-codified statutory rules karenge, aur final order hamesha ek authorized officer pass karega."

PART 2: THE CORE TECH (The 6-Stage Calibrated Pipeline)

 **Allocated Time: 1:15 – 3:20 (125 Seconds)**

 **Objective:** Explain the computer vision, geometry math, multilingual OCR, deterministic AST, and Section 63 BSA evidence vault with absolute scientific clarity.

[1:15 - 1:40] Eliminating LLM Hallucinations & The Pipeline Overview

(Stage Cue: Point to Pipeline diagram on slide)

*"Sabse pehle, ek critical architectural question: '**Kya AI hallucinations se false legal notices generate honge?**'"*

*Hamara jawab hai: **Katai nahi!***

Humne perception aur legal cognition ko completely decouple kiya hai. AI ka kaam sirf text polygons detect karna hai. Legal compliance hamara

***Deterministic Abstract Syntax Tree (AST) Rule Engine** execute karta hai.*

Exact statutory math — zero hallucinations!

Yeh hamara 6-stage end-to-end pipeline hai:"

[1:40 - 2:10] Stage 1 & 2: Optical Quality Gate & Planar Homography Calibration

(Stage Cue: Pull out a standard reference card / show card icon on slide)

[OPTION A - Direct Math & Caliper Truth]:

*"**Stage 1: Optical Quality Gate.** Kirana dukan ki tubelight glare aur motion blur ko hamara system 15 milliseconds mein Laplacian variance se filter karta hai.*

Frame clean hone par hi process aage badhta hai.

Stage 2: Spatial Metric Calibration. 2D photo mein pixels hote hain, millimeters nahi! Isliye hum standard **ISO/IEC 7810 reference card** ya ArUco marker use karte hain. **Planar Homography Matrix (H)** perspective tilt ko digitally un-tilt karta hai aur pixels ko exact real-world millimeters mein convert karta hai. Real physical test mein, digital vernier caliper truth **1.47 mm** thi — Nirikshak ne measure kiya **1.46 mm** — delta sirf **0.01 millimeters!**"

[OR OPTION B - Conceptual & Practical Clarity]:

"Stage 1 hamara **Quality Gate** hai, jo 15ms mein blur aur flash reflections reject karke inspector ko instant retake guide karta hai.

Stage 2 hamara **Geometric Calibration Engine** hai. Camera ke tilt aur distance ki wajah se pixels se real millimeters napna mathematically ill-posed problem hai. Hum ek standard reference card ke known dimensions se **3x3 Homography Matrix** solve karte hain. Isse smartphone camera ek verified digital micrometer ban jaata hai, jo Table-I font schedules ko sub-millimeter precision se verify karta hai."

[2:10 - 2:40] Stage 3 & 4: Multilingual Edge Vision & Cross-Facet Semantic Fusion

(Stage Cue: Emphasize 3D multi-surface packaging)

"Stage 3: Multilingual Edge Vision. Humne **DBNet++** for text detection aur **PP-OCRv4** for recognition use kiya hai. INT8 quantization ke through yeh standard CPU par bina kisi GPU ke English aur Devanagari Hindi text ko sirf **180 milliseconds** mein transcribe karta hai.

Stage 4: Cross-Facet Semantic Fusion. Real packages flat nahi hote, 3D hote hain! Front par brand hai, side flap par MRP aur batch, aur back par consumer care. Inspector multi-angle photos leta hai, aur hamara Fusion Engine spatial K-D proximity graph se saare tokens ko link karke ek composite 3D digital docket assemble karta hai — with complete source panel attribution."

[2:40 - 3:20] Stage 5 & 6: Statutory Rule AST & Section 63 BSA Evidentiary Vault

(Stage Cue: Show the dual screen: Rule Engine checks + Form-1 Notice PDF preview)

Stage 5: Statutory Rule Engine (AST). Rules 6, 9, 12 aur 24 of Packaged Commodities Rules 2011 code-level par mathematically enforce hote hain:

- Table-I PDP area ke according mandatory font height.
- Unit Sale Price calculation check — $\leq ₹0.02$ tolerance ke sath.
- Rule 12(b) metric symbol compliance — jaise m1 ke baad dot lagana illegal hai.

Stage 6: Section 63 BSA 2023 Evidentiary Vault.

Har raw camera buffer, device fingerprint, aur GPS timestamp ka **SHA-256 Merkle DAG Hash** generate hota hai. Yeh hash Form-1 Notice PDF aur Section 63 Certificate mein digitally seal hota hai. Agar photo ka ek pixel bhi alter kiya jaye, toh Merkle root invalidate ho jayegi. Courtroom mein evidence 100% tamper-evident hai!"

PART 3: FEASIBILITY, ECONOMIC VIABILITY & NATIONAL IMPACT

 **Allocated Time: 3:20 – 4:45 (85 Seconds)**

 **Objective:** Prove the system with the 4 real physical SKUs, demonstrate zero recurring GPU costs, show legal balance via Jan Vishwas Act, and present national scalability.

[3:20 - 3:55] Empirical Validation: 4 Real Retail SKUs Tested

(Stage Cue: Direct judges' attention to the Real Product Matrix table / slide)

"Humne koi synthetic test nahi kiya — humne market se 4 physical products purchase karke 38 high-res photographs par system validate kiya hai:

1. **Titan Fastrack Watch:** Clean PASS. Single-unit package hone ki wajah se Rule 6(1)(da) ke tehat Unit Sale Price exempt tha — system ne exemption recognize karke pass kiya.
2. **Himalaya Brahmi Tablets:** FAIL on technical deficit. Batch stamp font 1.46mm nikla bajaye 2.0mm ke.

3. **Boult Audio Earbuds:** FAIL on thermal sticker font deficit (1.24mm vs 2.0mm).
4. **Gopi Baba Hair Oil:** FAIL on 3 substantive violations: non-standard unit **100 ml** with dot, missing 'inclusive of all taxes' clause, aur missing consumer care email — jisme Section 48 ke tehat **₹25,000 Civil Compounding Notice** issue hua."*

[3:55 - 4:25] Economic Viability & Jan Vishwas Proportional Justice

(Stage Cue: Warm, reassuring tone — emphasizing Ease of Doing Business)

[OPTION A - Government Fiscal & Business Balance]:

"Ab baat karte hain **Economic Viability** ki:

Pehla: ₹0 Recurring Cloud GPU Expense. Saare models CPU-optimized ONNX format mein hain. System inspector ke existing phone ya field laptop par **Mode B (Offline SQLite)** mein bina internet ke chalta hai.

Doosra: Jan Vishwas Act 2023 Alignment. Packaging offenses ab decriminalize ho chuki hain. Nirikshak harassment rokta hai: Himalaya jaise minor font deficit par **15-Day Statutory Improvement Notice (₹0 fine)** deta hai taaki business rectify kar sake, jabki deliberate violations par compounding enforce karta hai.

Teesri: Pre-Compliance Simulator. Manufacturers carton print karane se pehle label design audit karke karodon ke product recall aur packaging waste se bach sakte hain."

[OR OPTION B - Operational Throughput & Enforcement Focus]:

*"Department ke liye iski **Viability teen pillars** par tiki hai:

1. **Throughput:** Per-package time 20 minute se ghata kar under 2 minute karta hai, jisse ek officer ki daily inspection capacity **15 se badh kar 200+ packages** ho sakti hai — a 13x surveillance expansion!
2. **Zero Cloud Infrastructure Burden:** INT8 quantization ki wajah se high-end GPU servers ki bilkul zaroorat nahi hai.
3. **Proportional Justice:** Minor printing defects par 15 din ka rectification notice, aur substantive violations par automated Section 48 civil compounding — ensuring true Ease of Doing Business."*

[4:25 - 4:45] Phased National Rollout & Readiness

(Stage Cue: Confident posture, ready for conclusion)

*"Deployment ke liye hamara phased roadmap ready hai:

- **Current State:** Fully working React web portal, FastAPI backend with 562 automated passing tests, deployed live at Oracle Cloud VPS.
- **Phase 1:** 50 District Headquarters mein field officers ke sath pilot deployment.
- **Phase 2:** Central DoCA eMaap portal ke sath standardized JSON schema integration, jiska data adapter hum already build kar chuke hain.
- **Phase 3:** Biometric CDAC Aadhaar e-Sign aur Native Android CameraX integration."*

CLOSING PUNCHLINE & HANDOFF

 **Allocated Time: 4:45 – 5:00 (15 Seconds)**

 **Objective:** Memorable, patriotic, professionally humble closing that transitions seamlessly into the Live Demo / Q&A.

[OPTION A - High Impact Closing]:

*"Respected Judges, NIRIKSHAK legal metrology inspection ko slow manual friction aur courtroom disputes se nikaal kar **scientific precision, transparent adjudication, aur digital sovereignty** ki taraf le jaata hai.*

Thank you, and we are now eager to demonstrate the live system!"

[OR OPTION B - Mission-Centric Closing]:

"NIRIKSHAK empowers our officers, protects 140 Crore Indian consumers, and safeguards honest businesses with verifiable law.

'AI Observes, Deterministic Rules Verify, Officer Decides.'

Thank you. We welcome your questions and invite you to our live demonstration."



JUDGES CROSS-EXAMINATION

DEFENSE CHEAT SHEET (Ready Answers)

Aapke presentation ke baad judges yeh 5 specific questions zaroor puchenge. Inke exact answers yaad rakhein:

Q1. "Aapne direct GPT-4 ya Gemini Vision API kyu nahi call kar li label check karne ke liye?"

Answer: "Sir, Generative LLMs probabilistic aur non-deterministic hote hain — wo hallucinate karte hain. Ek hi photo par LLM do alag verdicts de sakta hai. Quasi-judicial notices aur ₹25,000 ke penalties kisi black-box AI guess par issue nahi kiye ja sakte. Isliye humne Deep Learning ko sirf OCR tak restrict kiya, jabki compliance 100% deterministic code (AST Rule Engine) se evaluate hoti hai jo exact Gazette clauses follow karti hai."

Q2. "Photo tilt hone par font size galat nahi aayega?"

Answer: "Bilkul aa sakta hai agar linear scaling use ki jaye. Isliye hum **Planar Homography** calculate karte hain reference card ke 4 corners se. Yeh projective distortion ko mathematically un-tilt karta hai. Physical caliper test par 1.47mm ke mukable hamara measured error sirf 0.01mm tha."

Q3. "Remote rural mandi mein internet nahi hua toh?"

Answer: "Nirikshak mein **Mode B: Offline Field Mode** built-in hai. Localhost par embedded INT8 ONNX models aur local encrypted SQLite database chalta hai. Inspector bina internet ke inspection complete karta hai, aur connectivity aane par signed Merkle dockets cloud se sync ho jaate hain."

Q4. "Titan Watch par Unit Sale Price nahi tha, fir bhi pass kaise ho gayi?"

Answer: *"Sir, Rule 6(1)(da) ke 2nd Proviso (G.S.R. 779(E)) ke tehat agar Net Quantity exactly '1 Number' ya '1 Unit' hai, toh Unit Sale Price declare karna statutorily exempt hai! Generic AI isko fail kar deti, lekin hamara engine gazette exemptions ki exact legal depth janta hai."*

Q5. "Kya eMaap portal ke sath live integration complete hai?"

Answer: *"Sir, Department of Consumer Affairs ne abhi public third-party REST API expose nahi ki hai, isliye live direct webhook production mein DoCA gateway open hone par connect hoga. Lekin NIRIKSHAK ne complete standardized eMaap-compliant JSON schema aur adapter export build kar liya hai — ready for immediate handshake."*

Verified & Formatted for SIH26034 Team CYPHER / NIRIKSHAK